

Interpreting Data



Curriculum Ready

ACMSP: 248, 249, 250, 278, 283



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INTERPRETING DATA

Different lists of data have different properties. This unit is focused on the results and conclusions that can be found from these different properties.



Answer these questions, *before* working through the chapter.

I used to think:

The median of a data set is the middle score. Does this mean that the number of scores greater than the median is the same as the number of scores less than the median?

If the median splits the data in two halves, what do you think "quartiles" do?

The "range" of data is the difference between the highest and lowest score. What is "interquartile range"?

Answer these questions, *after* working through the chapter.

But now I think:

The median of a data set is the middle score. Does this mean that the number of scores greater than the median is the same as the number of scores less than the median?

If the median splits the data in two halves, what do you think "quartiles" do?

The "range" of data is the difference between the highest and lowest score. What is "interquartile range"?



What do I know now that I didn't know before?

Basic Statistics

Data is just a list of numbers called 'scores' or 'results'. The basic statistics that can be found from these scores are the **mean**, **median** or **mode**. (These are also called "measures of central tendency")

- The **mean** is the average score. The symbol for the **mean** is $\bar{x}$. It is found using the formula $\bar{x} = \frac{\sum fx}{\sum f}$.
- The **mode** is the score with the highest frequency. This is the score that occurs the most often.
- The **median** is the middle score when the scores are arranged in ascending order.
- The **cumulative frequency (cf)** is the sum of the frequencies for all scores less than or equal to that score.

Also remember that the symbol $\sum$ (called "sigma") means 'sum of' and so when $\sum x$ is written, it means the 'sum of the scores'.

Here is an example.

A group of people's height was measured (in cm) and the results were written in this table

Score (x)	Frequency (f)	Cumulative frequency (cf)	fx ← $f \times x$
110	3	3	$3 \times 110 = 330$
112	5	$3 + 5 = 8$	$5 \times 112 = 560$
113	10	$8 + 10 = 18$	$10 \times 113 = 1130$
115	9	$18 + 9 = 27$	$9 \times 115 = 1035$
116	8	$27 + 8 = 35$	$8 \times 116 = 928$
$\sum f$ means 'sum of frequencies' → $\sum f = 35$			$\sum fx = 3983$

- a How many people had a height of 112 cm?

The frequency of 112 is 5. This means there were 5 people who are 112 cm tall.

- b What is the cumulative frequency of 113 cm?

The cumulative frequency of 113 is the sum of the frequencies for scores of 113 or less.

The cf of 113 is $3 + 5 + 10 = 18$

- c Find the mean height, $\bar{x}$.

$$\begin{aligned}\bar{x} &= \frac{\sum fx}{\sum f} \\ &= \frac{3983}{35} = 113.8 \text{ cm}\end{aligned}$$

- d Find the mode.

The mode is 113 since it has the highest frequency (10).

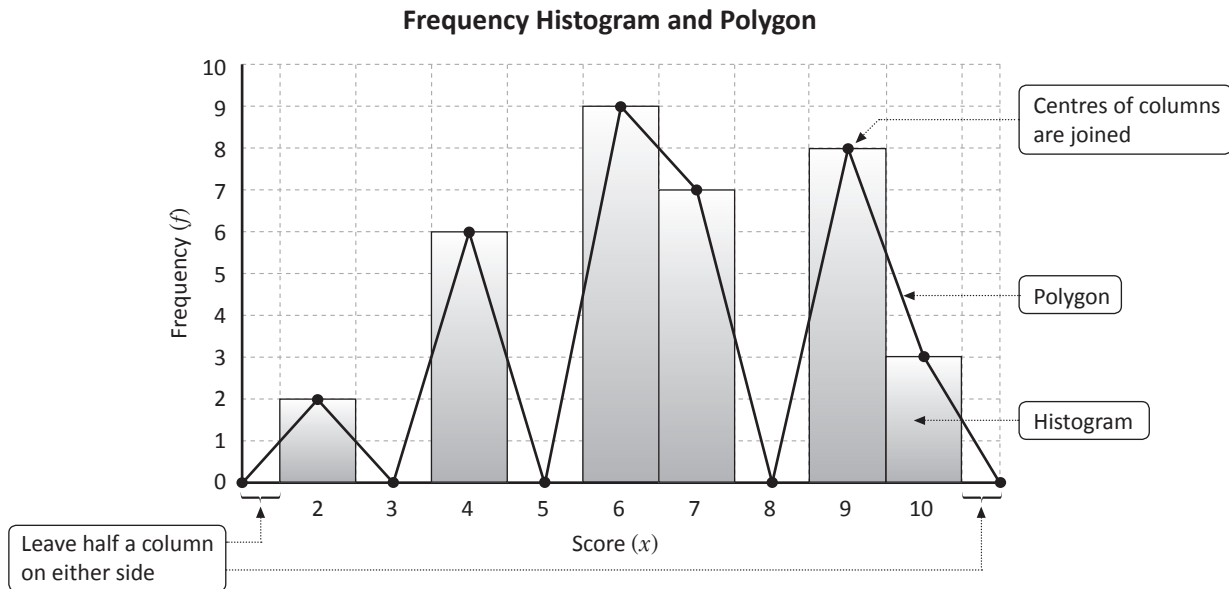
- e Find the median.

The median is the score in the middle. Since there are 35 scores the middle position is the 18th score.
 $\therefore$ the median is 113 since it would be in the 18th position (from the cf).

Frequency Histograms and Polygons

A histogram is a column graph based on data. A polygon is made up of straight lines joining the centres of these columns.

Answer the questions about this diagram



- a What is the frequency of the score 6?

9 (from the histogram)

- b What is the cumulative frequency of scoring a 6?

$$\begin{aligned}
 cf \text{ of } 6 &= \text{frequency of } 2 + \text{frequency of } 4 + \text{frequency of } 6 \\
 &= 2 + 6 + 9 \\
 &= 17
 \end{aligned}$$

- c How many scores are there?

$$\begin{aligned}
 \sum f &= 2 + 6 + 9 + 7 + 8 + 3 \\
 &= 35
 \end{aligned}$$

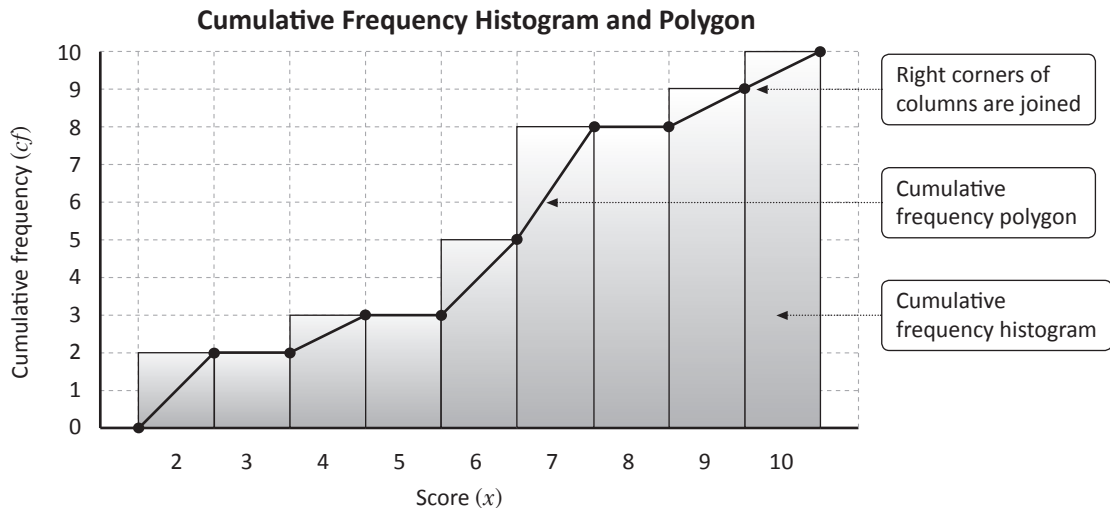
- d Find the mean (to 1 decimal place).

$$\begin{aligned}
 \bar{x} &= \frac{\sum fx}{\sum f} \\
 &= \frac{2 \times 2 + 6 \times 4 + 9 \times 6 + 7 \times 7 + 8 \times 9 + 3 \times 10}{35} \\
 &= 6.7 \text{ (1 d.p.)}
 \end{aligned}$$

Cumulative Frequency Histograms and Polygons

A cumulative frequency histogram can also be joined. The polygon joins the right corners of the histogram. A cumulative frequency polygon is also called an "ogive".

Answer these questions about the diagram below



- a What is the cumulative frequency of the score 6?

5 (from the histogram)

- b What is the frequency of scoring a 6?

$$\begin{aligned}
 \text{frequency of 6} &= \text{cumulative frequency of 6} - \text{cumulative frequency of 5} \\
 &= 5 - 3 \\
 &= 2
 \end{aligned}$$

- c How many scores are there?

Take the cumulative frequency of the last score. This means that there are 10 scores in total.

- d Find the median.

Since there were 10 scores, the median is the average of the two middle scores (in position 5 and 6)

$$\begin{aligned}
 \therefore \text{median} &= \frac{\text{score in 5}^{\text{th}} \text{ position} + \text{score in 6}^{\text{th}} \text{ position}}{2} \\
 &= \frac{6 + 7}{2} \\
 &= 6.5
 \end{aligned}$$

1. A group of people were asked how many languages they speak and this table was partly completed.

a Complete the table.

Number of languages (x)	Frequency (f)	fx	Cumulative frequency (cf)
1	20	$20 \times 1 = 20$	
2			38
3		$12 \times 3 = 36$	50
4	7		57
5	3		
	$\sum f = 60$	$\sum fx =$	

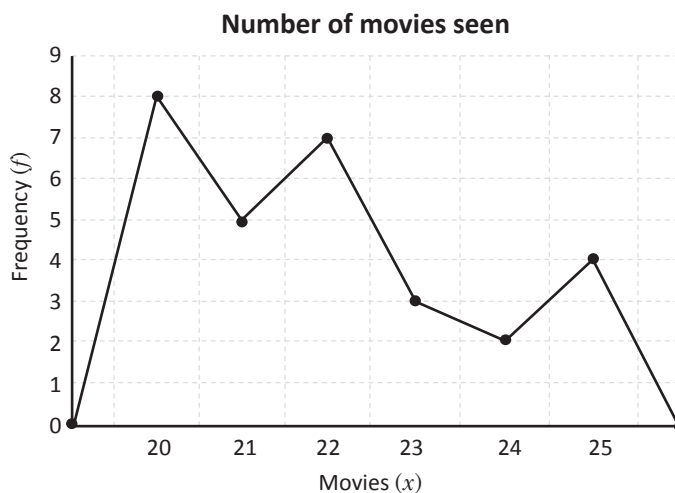
b Show that the mean is $\bar{x} = 2.25$.

c What is the median?

d What is the mode?

e Are $\sum f$ and the cumulative frequency of $x = 5$ equal? Why?

2. A group of people were asked how many movies they had seen in the last year. The diagram below shows the frequency polygon for the results.



- a Use the diagram to complete the table below.

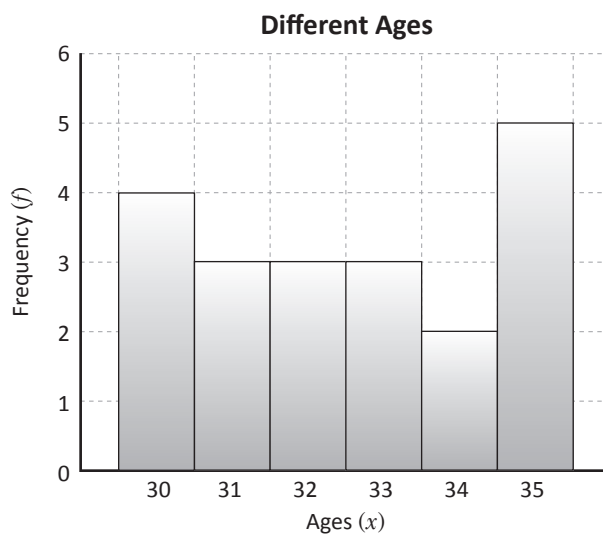
Movies (x)	Frequency (f)	fx	Cumulative frequency (cf)
$\sum f =$		$\sum fx =$	

- b What is the mean (to 2 decimal places if necessary)?

- c What is the median?

- d What is the mode?

3. A group of people were asked their age and this frequency histogram was produced.



a Complete the table below.

Ages (x)	Frequency (f)	fx	Cumulative frequency (cf)
	$\sum f =$	$\sum fx =$	

b Complete the polygon on the diagram.

c Find the mean age (to 2 d.p. if necessary).

d What is the mode?

e What is the median?

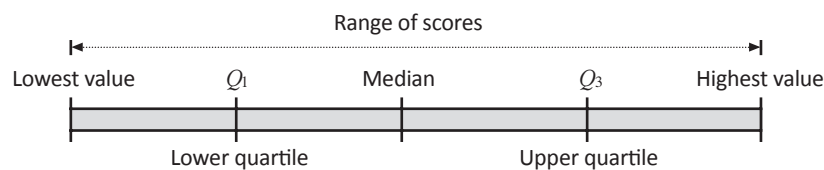
The median is the middle score of the data (or the average of the two middle scores). This means that 50% of the scores are less than or equal to the median. Quartiles work the same way.

Quartiles

There are 3 quartiles.

- The first quartile – written as Q_1 – is the score that 25% of the scores are less than or equal to. Q_1 is the median of the lower half of the scores.
- The second quartile – written as Q_2 – is the median.
- The third quartile – written as Q_3 – is the score that 75% of the scores are less than or equal to. Q_3 is the median of the upper half of the scores.

In other words the **quartiles** divide the data into **quarters**.



Answer the following questions about this table of data

Score (x)	Frequency (f)	Cumulative frequency (cf)
1	3	3
2	4	7
3	6	13
4	2	15
5	1	16
6	4	20
$\sum f = 20$		

- a How many scores are there in total?

There are 20 scores since $\sum f = 20$

- b What is the median (Q_2)?

The median will be the average of the two middle scores. That is the average of the scores in 10th and 11th position. Since the cf of $x = 2$ is 7 and the cf of $x = 3$ is 13, so the scores in 10th and 11th position are both 3.

$$\therefore Q_2 = \frac{3 + 3}{2} = 3$$

- c Find the lower quartile Q_1

Since there are 20 scores, the Q_1 will be the average of the scores in the 5th and 6th positions. From the table, the score in 5th position is 2, and the score in the 6th position is 2.

$$\therefore Q_1 = \frac{2 + 2}{2} = 2$$

- d Find the upper quartile Q_3 ?

Since there are 20 scores, the Q_3 will be the average of the scores in the 15th and 16th positions. From the table, the score in the 15th position is 4, and the score in the 16th position is 5.

$$\therefore Q_3 = \frac{4 + 5}{2} = 4.5$$

Range and Interquartile Range

- The range of a data set is the difference between the highest score and the lowest score.
- The **interquartile range** is the difference between the upper and lower quartiles. It is written as IQR .

$$\therefore IQR = Q_3 - Q_1$$

A 5-point summary of a data set is a list of : The lowest value; Q_1 ; Q_2 ; Q_3 and the highest value. Here is an example:

Answer the questions about the table below

Score (x)	Frequency (f)	Cumulative frequency (cf)
10	5	5
12	8	13
14	4	17
16	10	27
18	6	33
20	3	36
$\sum f = 36$		

a Find Q_1

There are 36 scores in total, so Q_1 is the average of the 9th and 10th scores (median of the lower half). The cf of $x = 10$ is 5 and the cf of $x = 12$ is 13.

$$\therefore Q_1 = \frac{12 + 12}{2} = 12$$

b Find Q_2

There are 36 scores in total, so the median is the average of the scores in 18th and 19th position:

$$\therefore Q_2 = \frac{16 + 16}{2} = 16$$

c Find Q_3

There are 36 scores in total, so Q_3 is the average of the in 27th and the 28th scores. The cf of $x = 16$ is 27 and cf of $x = 18$ is 33:

$$\therefore Q_3 = \frac{16 + 18}{2} = 17$$

d Find the range and interquartile range?

$$\begin{aligned} \text{Range} &= \text{Highest Score} - \text{Lowest Score} \\ &= 20 - 10 = 10 \end{aligned}$$

$$\begin{aligned} IQR &= Q_3 - Q_1 \\ &= 17 - 12 = 5 \end{aligned}$$

e Write down a 5-point summary for this set of data

- Lowest score = 10
- $Q_1 = 12$ (lower quartile)
- $Q_2 = 16$ (the median)
- $Q_3 = 17$ (upper quartile)
- Highest score = 20

1. Here is a list of data: 5 , 7 , 2 , 3 , 8 , 0 , 1 , 2 , 4 , 8 , 4 , 2 , 7 , 8 , 0 , 1.

- a Arrange this data into ascending order.
- b Find the median of this data set.
- c Find Q_1 .
- d Find the upper quartile.
- e Write down a 5-point summary of this data set.
- f What is the range of this data set?
- g What is in the interquartile range of this data set?

2. A group of students' exam results are displayed in the table below.

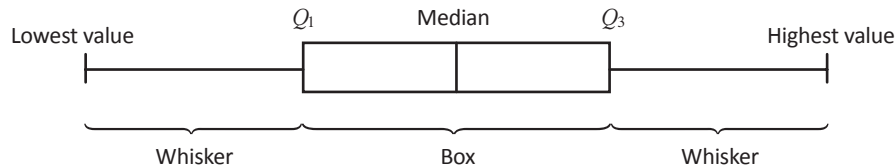
Results in % (x)	Number of students (f)	Cumulative frequency (cf)
10	5	
20	3	
30	1	
40	8	
50	3	
60	9	
70	6	
80	3	
90	8	
100	6	
$\sum f = 52$		

- Complete the cumulative frequency column.
- How many students were used to create the table?
- In which position is the median? What is the median?
- Find Q_1 and Q_3 .
- Find the interquartile range.
- Write a 5-point summary on this data.

Box-and-Whisker Plots

A 5-point summary is used to plot a "Box-and-Whisker" plot for a set of Data.

These are used to compare different data sets. They are drawn like this:



They are always drawn against a number line.

Answer these questions about this set of data

29, 26, 30, 22, 30, 21, 22, 22, 25, 24, 21, 26

- a Arrange this data into ascending order:

21, 21, 22, 22, 22, 24, 25, 26, 26, 29, 30, 30

- b Find a 5-point summary.

Q_1 Q_2 Q_3

There are 12 numbers in the data set.

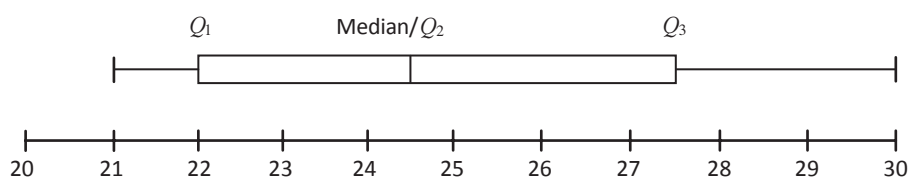
- The lowest number is 21.
- Q_1 will be the average of the 3rd and the 4th position so:

$$\therefore Q_1 = \frac{22 + 22}{2} = 22$$
- The median is the average of the numbers in 6th and 7th position.

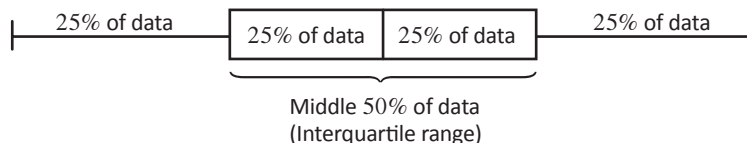
$$\therefore Q_2 = \frac{24 + 25}{2} = 24.5$$
- Q_3 will be the average of the 9th and the 10th position so:

$$\therefore Q_3 = \frac{26 + 29}{2} = 27.5$$
- The greatest number in the set is 30.

- c Draw a box-and-whisker plot for this set of data.



In any Box-and-Whisker plot the diagram represents:



Box-and-Whisker plots are used to compare different sets of data.

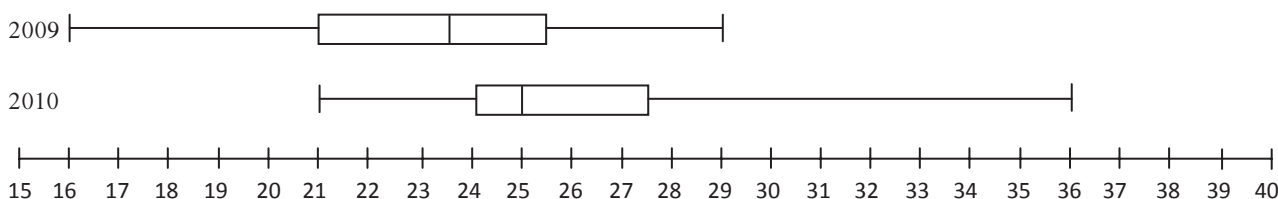
The table below shows average temperatures for a city over two years

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
2009	28	22	24	22	16	26	20	25	29	20	23	24
2010	21	24	26	25	24	25	24	25	30	27	28	36

- a Find a 5-point summary for the temperatures in 2009 and 2010.

	2009	2010
Ascending Order	16, 20, 20, 22, 22, 23, 24, 24, 25, 26, 28, 29	21, 24, 24, 24, 25, 25, 25, 26, 27, 28, 30, 36
Lowest Temperature	16	21
Q_1	21	24
Q_2	23.5	25
Q_3	25.5	27.5
Highest Temperature	29	36

- b Draw box and whisker plots for the average temperatures in 2009 and 2010.



- c Which year had the greater range of temperature?

It can be seen that the box-and-whisker plot for 2010 is longer than that of 2009. This means that 2010 had the greater range in temperature.

- d Which year had the greater interquartile range?

It can be seen that the box-and-whisker plot for 2009 has a longer interquartile range (the 'box' part) than 2010. This means 2009 had a greater interquartile range in temperature.

Standard Deviation

Standard deviation measures the average distance each score is away from the mean. It has this symbol using lower case sigma σ_n , pronounced 'sigma-n'. This is the formula for σ_n :

$$\sigma_n = \sqrt{\frac{\sum (x - \bar{x})^2}{n}}$$

Where $\bar{x}$ is the mean and $\sum$ still means 'sum of'. Here is an example:

Find the standard deviation (correct to 1 decimal place) of this set of data: 11, 8, 13, 3, 9, 15, 17, 17, 6, 11

- Find the mean, $\bar{x}$:

$$\begin{aligned}\bar{x} &= \frac{\text{sum of scores}}{n} \\ &= \frac{110}{10} \\ &= 11\end{aligned}$$

- Draw a table with these 3 columns.

This is called "mean difference"

"mean difference" is squared

Score (x)	$x - \bar{x}$	$(x - \bar{x})^2$
11	$11 - 11 = 0$	0
8	$8 - 11 = -3$	9
13	$13 - 11 = 2$	4
3	$3 - 11 = -8$	64
9	$9 - 11 = -2$	4
15	$15 - 11 = 4$	16
17	$17 - 11 = 6$	36
17	$17 - 11 = 6$	36
6	$6 - 11 = -5$	25
11	$11 - 11 = 0$	0
$\sum (x - \bar{x}) = 0$		$\sum (x - \bar{x})^2 = 194$

This total will ALWAYS be zero.
If not, a mistake has been made.

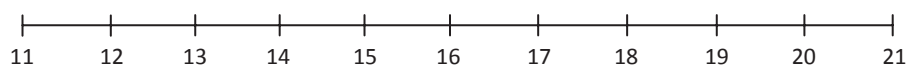
- Use the formula for standard deviation.

$$\begin{aligned}\sigma_n &= \sqrt{\frac{\sum (x - \bar{x})^2}{n}} \\ &= \sqrt{\frac{194}{10}} \\ &= 4.4 \text{ (1 d.p.)}\end{aligned}$$

1. An athlete runs the same race 16 times. This is how long it takes him (in seconds) to run each time:

14 , 12 , 18 , 14 , 16 , 18 , 19 , 14 , 16 , 17 , 15 , 13 , 20 , 16 , 14 , 19

- a Write these times in ascending order.
- b Find a 5-point summary of this data.
- c Draw a box-and-whisker plot for this data.



- d What is the range of the data?
- e What is the interquartile range of the data?

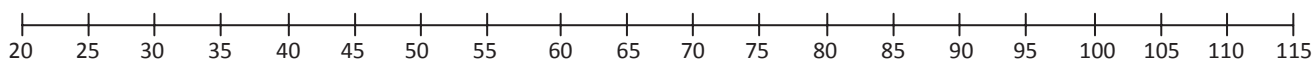
2. During 8 days, a cat and a dog eat an amount of food (in grams) according to the table below.

	Mon	Tues	Wed	Thur	Fri	Sat	Sun	Mon
Cat	70	100	40	90	50	70	55	100
Dog	65	100	90	80	75	85	50	85

a Arrange each set of data into ascending order

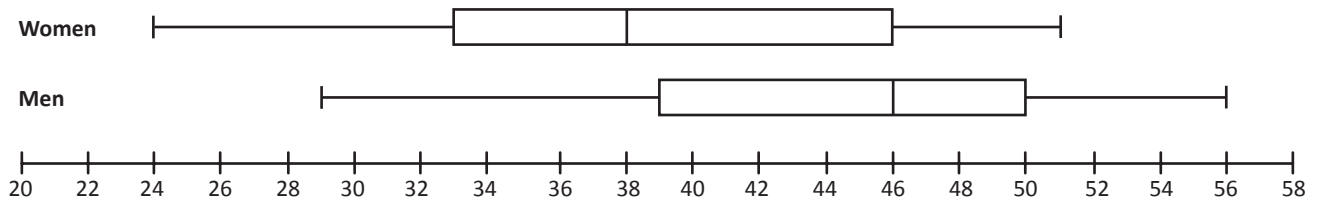
b Find a 5-point summary for each data set.

c Draw box-and-whisker plots for the different data sets on the number line below.



d From the box-and-whisker plot, which had the greater interquartile range? Find the interquartile range.

3. 150 men and 150 women were in a survey and these are the resulting box-and-whisker plots from their ages:



- How old was the youngest woman in the survey?
- How old was the oldest man in the survey?
- What is the median age for the women?
- Find Q_1 for the men.
- Find Q_3 for the women.
- What is the age that half the men were older than?

4. Ava counted the number of books she read each month for a year. She wrote them in the table below:

Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
5	6	4	1	6	1	9	9	10	10	5	6

a Find $\bar{x}$, the mean.

b Complete the table below for the above set of data.

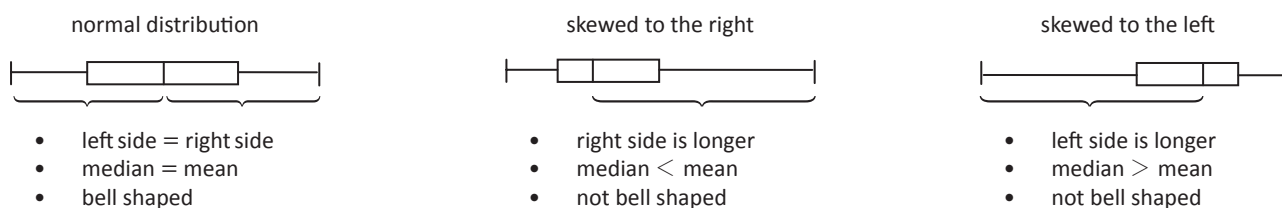
x	$x - \bar{x}$	$(x - \bar{x})^2$
5	-1	
6	0	
4		
1	-5	
6		
1	-5	
9		
9	3	9
10	4	
10		
5		
6	0	0
$\sum (x - \bar{x}) =$		$\sum (x - \bar{x})^2 =$

c What is the formula to calculate σ_n ?

d Show that the standard deviation of the above set of data to 2 decimal places is 2.97.

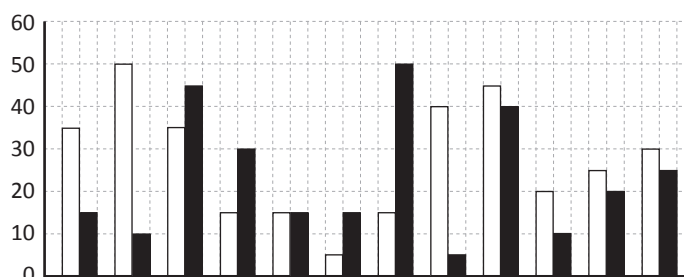
Skewness of Data

A set of data can be one of three things: a normal distribution, skewed to the right or skewed to the left.



This is only a general rule of thumb which holds most of the time and exceptions to this rule do occur. This can be used to compare different data sets.

Two data sets are shown on the column graph below, data set 1 (white) and data set 2 (black).



a Write both data sets in ascending order

Data Set 1 (white): 5, 15, 15, 15, 20, 25, 30, 35, 35, 40, 45, 50

Data Set 2 (black): 5, 10, 10, 15, 15, 15, 20, 25, 30, 40, 45, 50

b Find the median of both data sets

Data Set 1:

$$\begin{aligned}\text{median} &= \frac{25 + 30}{2} \\ &= 27.5\end{aligned}$$

Data Set 2:

$$\begin{aligned}\text{median} &= \frac{15 + 20}{2} \\ &= 17.5\end{aligned}$$

c Find the mean of both data sets and comment on the skewness of each data set.

Data set 1:

$$\text{Mean} = 27.5$$

This is a normal distribution since mean = median

Data set 2:

$$\text{Mean} = 23\frac{1}{3}$$

This is skewed to the right since median < mean

Spread of Data

The *spread* of a data set measures how consistent (close to the mean) a data set is. This depends on:

- Range – the wider the range of the data set the less likely scores will be close to the mean
- Interquartile range – the wider the range of the data set the less likely scores will be close to the mean
- Standard deviation – σ_n measures how far the scores are from the mean.

A gameplayer tries two strategies for playing a game. He tries each strategy eight times and these are the points received

Strategy 1: 34, 35, 28, 28, 30, 31, 32, 27

Strategy 2: 1, 13, 5, 10, 16, 14, 1, 5

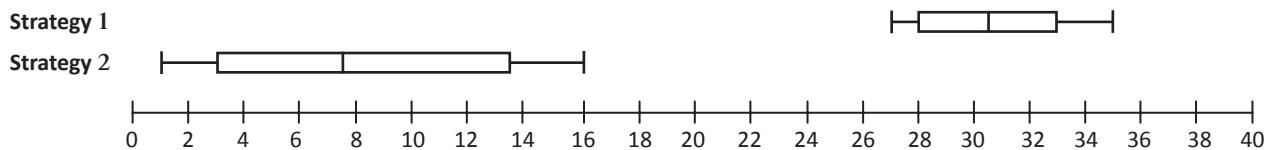
a Find a 5-point summary for each strategy.

	Strategy 1	Strategy 2
Lowest	27	1
Q_1	28	$\frac{1+5}{2} = 3$
Q_2	30.5	7.5
Q_3	33	$\frac{13+14}{2} = 13.5$
Highest	35	16

b Find the range and interquartile range for each strategy.

	Strategy 1	Strategy 2
Range	$35 - 27 = 8$	$16 - 1 = 15$
IQR	$33 - 28 = 5$	$13.5 - 3 = 10.5$

c Draw a box-and-whisker plot for each strategy.



d Which strategy do you expect to have a larger standard deviation?

Strategy 2 has a larger range and interquartile range, so it is expected to have a larger standard deviation.

e Find the standard deviation of both strategies to 2 decimal places.

$$\sigma_n = \sqrt{\frac{\sum (x - \bar{x})^2}{n}}$$

σ_n for Strategy 1: 2.74 (2 d.p.)
 σ_n for Strategy 2: 5.53 (2 d.p.)

As expected, the standard deviation for Strategy 2 is greater.

1. Two movies received reviews from eight critics who gave the movie a score between 1 and 10. Here are their results:

Movie 1: 3, 8, 8, 6, 3, 5, 2, 5

Movie 2: 8, 7, 3, 4, 7, 10, 6, 3

a Use this table to find the standard deviation of the Movies' scores:

Movie 1		
Score (x)	$x - \bar{x}$	$(x - \bar{x})^2$
3		
8		
8		
6		
3		
5		
2		
5		
$\sum(x - \bar{x}) =$		$\sum(x - \bar{x})^2 =$

Movie 2		
Score (x)	$x - \bar{x}$	$(x - \bar{x})^2$
8		
7		
3		
4		
7		
10		
6		
3		
$\sum(x - \bar{x}) =$		$\sum(x - \bar{x})^2 =$

b Complete these tables for the movies:

	Movie 1	Movie 2
Lowest		
Q_1		
Q_2		
Q_3		
Highest		
σ_n		

c How are the scores for each movie skewed?

d Which movie has the more consistent scores?

2. At the olympics, divers receive a score between 1 and 10 each time they dive. These are the scores after 12 dives for the divers who came in first and second place.

Diver A

7, 8, 5, 7, 8, 6, 6, 5, 8, 5, 8, 5

Diver B

7, 6, 4, 5, 6, 5, 10, 9, 8, 6, 9, 9

- a Find a 5-point summary for each diver's scores
- b Find the range and interquartile range of each diver's scores
- c Draw a box-and-whisker plot for each of the divers' scores. Are the scores skewed?



- d Draw up a table with the headings, **score (x)**, $x - \bar{x}$ and $(x - \bar{x})^2$ and use it to find σ_n for both divers.

- e If the winner is based on the total scores, which diver won?

- f Which diver had more consistent scores?

3. Answer these questions about skewness of data:

- a If the median is less than the mean which way would the data be skewed (according to the rule of thumb)?

- b When is data skewed to the left (according to the rule of thumb)?

- c Sketch a box-and-whisker plot representing data that is skewed to the left.

- d Standard deviation is a measure of how far each score is from the mean. What does this mean?

- e How is the standard deviation affected by the consistency of the scores?

- f Does data with a higher or lower standard deviation have more consistency?

Basics:

1. a

Number of languages (x)	Frequency (f)	fx	Cumulative frequency (cf)
1	20	$20 \times 1 = 20$	$0 + 20 = 20$
2	$38 - 20 = 18$	$18 \times 2 = 36$	$20 + 18 = 38$
3	$50 - 38 = 12$	$12 \times 3 = 36$	$38 + 12 = 50$
4	7	$4 \times 7 = 28$	$50 + 7 = 57$
5	3	$5 \times 3 = 15$	$57 + 3 = 60$
$\sum f = 60$		$\sum fx = 20 + 36 + 36 + 28 + 15 = 135$	

b $\bar{x} = \frac{\sum fx}{\sum f} = \frac{135}{60} = 2.25$

c median = 2

d mode = 1

e Yes, $\sum f$ and the cumulative frequency of $x = 5$ are equal as $x = 5$ is the highest score in the data. This means all the scores are less than or equal to 5.

Basics:

2. a

Movies (x)	Frequency (f)	fx	Cumulative frequency (cf)
20	8	$20 \times 8 = 160$	8
21	5	$21 \times 5 = 105$	$8 + 5 = 13$
22	7	$22 \times 7 = 154$	$13 + 7 = 20$
23	3	$23 \times 3 = 69$	$20 + 3 = 23$
24	2	$24 \times 2 = 48$	$23 + 2 = 25$
25	4	$25 \times 4 = 100$	$25 + 4 = 29$
$\sum f = 29$		$\sum fx = 636$	

b $\bar{x} = 21.93$ (2 d.p.)

c median = 22

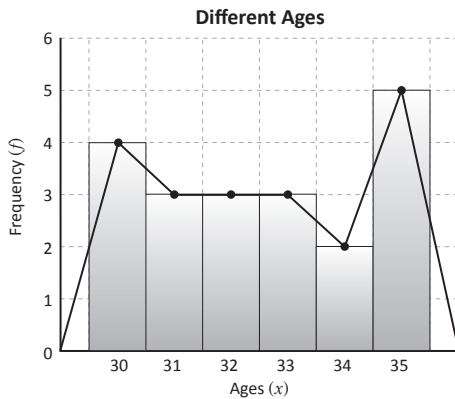
d mode = 20

3. a

Ages (x)	Frequency (f)	fx	Cumulative frequency (cf)
30	4	$30 \times 4 = 120$	4
31	3	$31 \times 3 = 93$	$4 + 3 = 7$
32	3	$32 \times 3 = 96$	$7 + 3 = 10$
33	3	$33 \times 3 = 99$	$10 + 3 = 13$
34	2	$34 \times 2 = 68$	$13 + 2 = 15$
35	5	$35 \times 5 = 175$	$15 + 5 = 20$
$\sum f = 20$		$\sum fx = 651$	

Basics:

3. b



- c $\bar{x} = 32.55$
- d mode = 35
- e median = 32.5

Knowing More:

1. a 0, 0, 1, 1, 2, 2, 2, 3, 4, 4, 5, 7, 7, 8, 8, 8
- b median = 3.5
- c $Q_1 = 1.5$
- d $Q_3 = 7$
- e
- Lowest score = 0
 - $Q_1 = 1.5$ (lower quartile)
 - $Q_2 = 3.5$ (the median)
 - $Q_3 = 7$ (upper quartile)
 - Highest score = 8
- f Range = 8
- g $IQR = 5.5$

Knowing More:

2. a

Results in % (x)	Number of students (f)	Cumulative frequency (cf)
10	5	5
20	3	$5 + 3 = 8$
30	1	$8 + 1 = 9$
40	8	$9 + 8 = 17$
50	3	$17 + 3 = 20$
60	9	$20 + 9 = 29$
70	6	$29 + 6 = 35$
80	3	$35 + 3 = 38$
90	8	$38 + 8 = 46$
100	6	$46 + 6 = 52$
$\sum f = 52$		

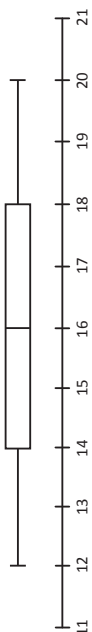
- b The total number of students used is 52
- c median = 60
- d $Q_1 = 40$
 $Q_3 = 90$
- e $IQR = 50$
- f
- Lowest score = 10
 - $Q_1 = 40$ (lower quartile)
 - $Q_2 = 60$ (the median)
 - $Q_3 = 90$ (upper quartile)
 - Highest score = 100

Using Our Knowledge:

1. a 12, 13, 14, 14, 14, 15, 16, 16, 16, 17, 18, 18, 19, 19, 20
- b
- Lowest score = 12
 - $Q_1 = 14$ (lower quartile)
 - $Q_2 = \frac{16 + 16}{2} = 16$ (the median)
 - $Q_3 = 18$ (upper quartile)
 - Highest score = 20

Using Our Knowledge:

1. c



d Range = 8

e $IQR = 4$

2. a

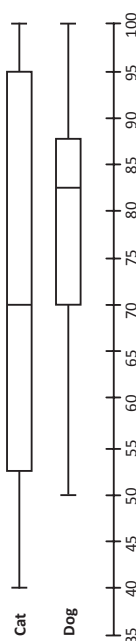
Cat	40	50	55	70	70	90	100	100
Dog	50	65	75	80	85	85	90	100

- b Cat:**
- Lowest score = 40
 - $Q_1 = 52.5$ (lower quartile)
 - $Q_2 = 70$ (the median)
 - $Q_3 = 95$ (upper quartile)
 - Highest score = 100

- Dog:**
- Lowest score = 50
 - $Q_1 = 70$ (lower quartile)
 - $Q_2 = 82.5$ (the median)
 - $Q_3 = 87.5$ (upper quartile)
 - Highest score = 100

Using Our Knowledge:

2. c



d $IQR(\text{cat}) = 42.5$

$IQR(\text{dog}) = 17.5$

3. a From the box-and-whisker plot the youngest woman was 24 years old.

b The oldest man was 56 years old

c The median age for the women was 38 years old

d Q_1 for the men was 39 years old

e Q_3 for the women was 46 years old

f The age that half the men were older than is the median. The median is 46 years old.

Using Our Knowledge:

4. a $\bar{x} = 6$

b

x	$x - \bar{x}$	$(x - \bar{x})^2$
5	-1	$(-1)^2 = 1$
6	0	0
4	$4 - 6 = -2$	4
1	-5	25
6	$6 - 6 = 0$	0
1	-5	25
9	$9 - 6 = 3$	9
9	3	9
10	4	16
10	4	16
5	-1	1
6	0	0
$\sum (x - \bar{x}) = 0$		$\sum (x - \bar{x})^2 = 106$

c $\sigma_n = \sqrt{\frac{\sum (x - \bar{x})^2}{n}}$

d $\sigma_n = \sqrt{\frac{\sum (x - \bar{x})^2}{n}} = \sqrt{\frac{106}{12}} = 2.97$

Thinking More:

1. a

Movie 1		
Score (x)	$x - \bar{x}$	$(x - \bar{x})^2$
3	$3 - 5 = -2$	4
8	$8 - 5 = 3$	9
8	$8 - 5 = 3$	9
6	$6 - 5 = 1$	1
3	$3 - 5 = -2$	4
5	$5 - 5 = 0$	0
2	$2 - 5 = -3$	9
5	$5 - 5 = 0$	0
$\sum (x - \bar{x}) = 0$		$\sum (x - \bar{x})^2 = 36$

Movie 1: $\sigma_n = 2.12$

Thinking More:

1. a

Movie 2		
Score (x)	$x - \bar{x}$	$(x - \bar{x})^2$
8	$8 - 6 = 2$	4
7	$7 - 6 = 1$	1
3	$3 - 6 = -3$	9
4	$4 - 6 = -2$	4
7	$7 - 6 = 1$	1
10	$10 - 6 = 4$	16
6	$6 - 6 = 0$	0
3	$3 - 6 = -3$	9
$\sum (x - \bar{x}) = 0$		$\sum (x - \bar{x})^2 = 44$

Movie 1: $\sigma_n = 2.35$

b

	Movie 1	Movie 2
Lowest	2	3
Q_1	3	3.5
Q_2	5	6.5
Q_3	7	7.5
Highest	8	10
σ_n	2.12	2.35

c For Movie 1:
the mean = the median = 5

$\therefore$ the data is not skewed and the scores are distributed normally

For Movie 2:
the mean = 6 and the median = 6.5,
median > mean

$\therefore$ the scores are skewed to the left (ie there are more scores that are less than the median than there are scores greater than the median)

Thinking More:

1. d

	Movie 1	Movie 2
Range	$8 - 2 = 6$	$10 - 3 = 7$
Interquartile range	$7 - 3 = 4$	$7.5 - 3.5 = 4$
σ_n	2.12	2.35

Movie 1 has the more consistent scores than Movie 2. This is because the standard deviation of the scores for Movie 1 is less than for Movie 2.

2. a

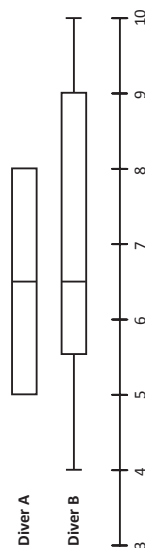
	Diver A	Diver B
Lowest	5	4
Q_1	5	5.5
Q_2	6.5	6.5
Q_3	8	9
Highest	8	10

b

	Diver A	Diver B
Range	$8 - 5 = 3$	$10 - 4 = 6$
Interquartile range	$8 - 5 = 3$	$9 - 5.5 = 3.5$

Thinking More:

2. c



The scores for Diver A are not skewed. The scores for Diver B are skewed to the right (the box-and-whisker plot is longer on the right hand side of the median)

d

Diver A		
x	$x - \bar{x}$	$(x - \bar{x})^2$
5	-1.5	2.25
5	-1.5	2.25
5	-1.5	2.25
5	-1.5	2.25
6	-0.5	0.25
6	-0.5	0.25
7	0.5	0.25
7	0.5	0.25
8	1.5	2.25
8	1.5	2.25
8	1.5	2.25
8	1.5	2.25
$\sum (x - \bar{x}) = 0$		$\sum (x - \bar{x})^2 = 19$

Diver A: $\sigma_n = 1.126$

Thinking More:

2. d

Diver B		
x	$x - \bar{x}$	$(x - \bar{x})^2$
4	-3	9
5	-2	4
5	-2	4
6	-1	1
6	-1	1
6	-1	1
7	0	0
8	1	1
9	2	4
9	2	4
9	2	4
10	3	9
$\sum (x - \bar{x}) = 0$		$\sum (x - \bar{x})^2 = 42$

Driver B: $\sigma_n = 1.87$

- e If the winner was based on total score, Diver B won.
- f Diver A had the more consistent scores. This is shown by the lower standard deviation of Diver A's scores.

3. a The data is skewed to the right.

- b When median $>$ mean

Thinking More:

3. c



- d The standard deviation measures how spread out the scores are. The larger a standard deviation is, the further the scores are from the mean.
- e The more consistent the scores, the lower the standard deviation. The less consistent the scores, the higher the standard deviation.
- f Data with a lower standard deviation has more consistency.



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